

Chapter 1

Getting Started

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Component List

Links are only for reference.

1. STM32 Blue Pill (STM32F103C8T6) (1 pcs) *****Important: Make sure to solder its pins from buyer** [\[Link\]](#)
2. ST Link V2 Programmer for STM32 (1 pcs) **[Make sure it has 4 female to female wires]** [\[Link\]](#)
3. 10K Resistor (10 pcs)
4. 220 Ω resistors (10 pcs)
5. Bread Board (1 pcs) **[Transparent breadboard is preferable.]** [\[Link\]](#)
6. Jumper Wire (Male to Male) (20 pcs)
7. Multimeter (1 pcs) **[Optional]**
8. 2 pin Pushbutton (5 pcs)
9. LED (5 pcs RED + 5 pcs GREEN + 5 pcs YELLOW)
10. 3 pin RGB LED (5 pcs)
11. 10k Ω potentiometer (3 pcs)
12. Photoresistor (2 pcs)
13. TMP36 Temperature Sensor (1 pc)

Software to be Installed

1. Arduino IDE (2.3.1) [I personally prefer 1.8.18]
2. Keil uVision IDE (5.39.0.0)
3. STM32 CubeIDE (1.15.0)

Arduino IDE Install

Version: 2.3.1 (Download only 2.3.1 version, not latest version)

Download Link: <https://github.com/arduino/arduino-ide/releases>

3 weeks ago

github-actions

2.3.1

c54fbc8

Compare

2.3.1

Fixed

- Debug view blank on first debugging session of a sketch (#2354)

Security

- Update msgpackr to resolve CVE-2023-52079 (#2360)
- Update axios to resolve CVE-2023-45857 (#2360)
- Update follow-redirects to resolve CVE-2023-26159 (#2360)

Full Changelog: [2.3.0...2.3.1](#)

Assets 14

arduino-ide_2.3.1_Linux_64bit.AppImage	181 MB	3 weeks ago
arduino-ide_2.3.1_Linux_64bit.zip	181 MB	3 weeks ago
arduino-ide_2.3.1_macOS_64bit.dmg	185 MB	3 weeks ago
arduino-ide_2.3.1_macOS_64bit.zip	181 MB	3 weeks ago
arduino-ide_2.3.1_macOS_arm64.dmg	175 MB	3 weeks ago
arduino-ide_2.3.1_macOS_arm64.zip	171 MB	3 weeks ago
arduino-ide_2.3.1_Windows_64bit.exe	143 MB	3 weeks ago
arduino-ide_2.3.1_Windows_64bit.msi	152 MB	3 weeks ago
arduino-ide_2.3.1_Windows_64bit.zip	185 MB	3 weeks ago

Keil IDE Install

Version: 5.39.0.0 (Fill up following form to get the Download Link)

Download Link: <https://www.keil.com/demo/eval/arm.htm>

Video Link: <https://www.youtube.com/watch?v=buQtJjgw2pE>

ST-Link driver Install

Download Link: <https://www.st.com/en/development-tools/stsw-link009.html>

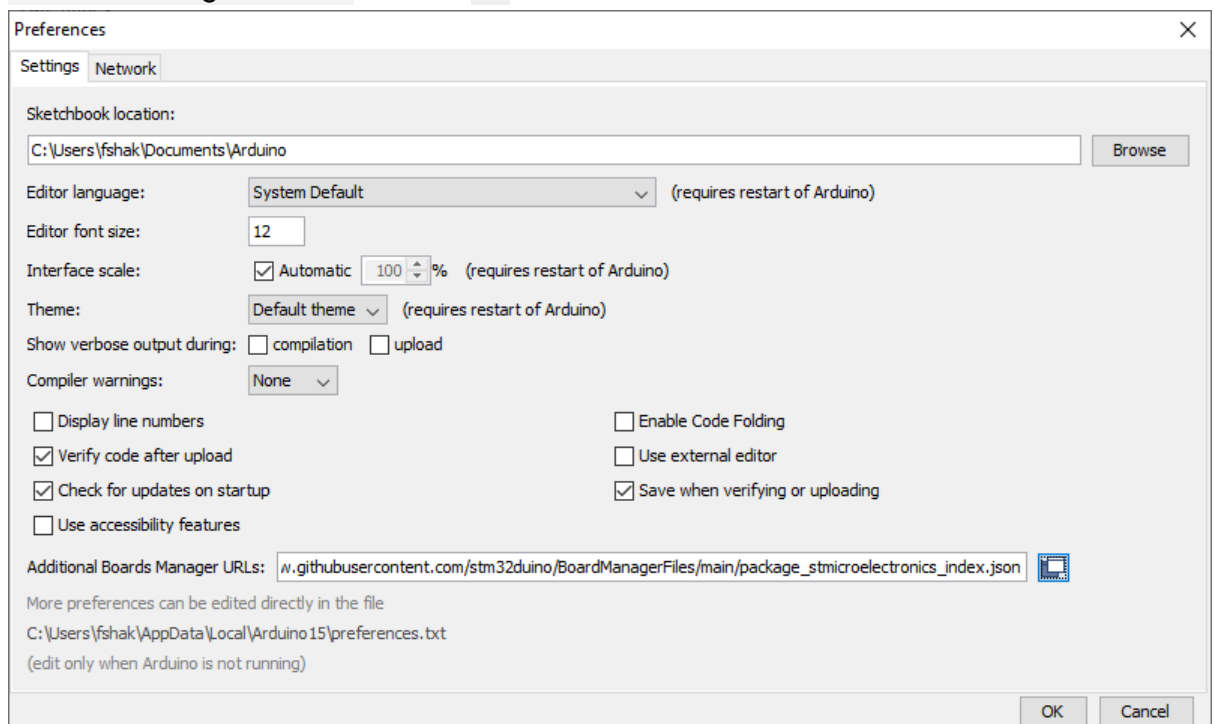
STM32 Cube Programmer Install

Download Link:
<https://www.st.com/en/development-tools/stm32cubeprog.html>

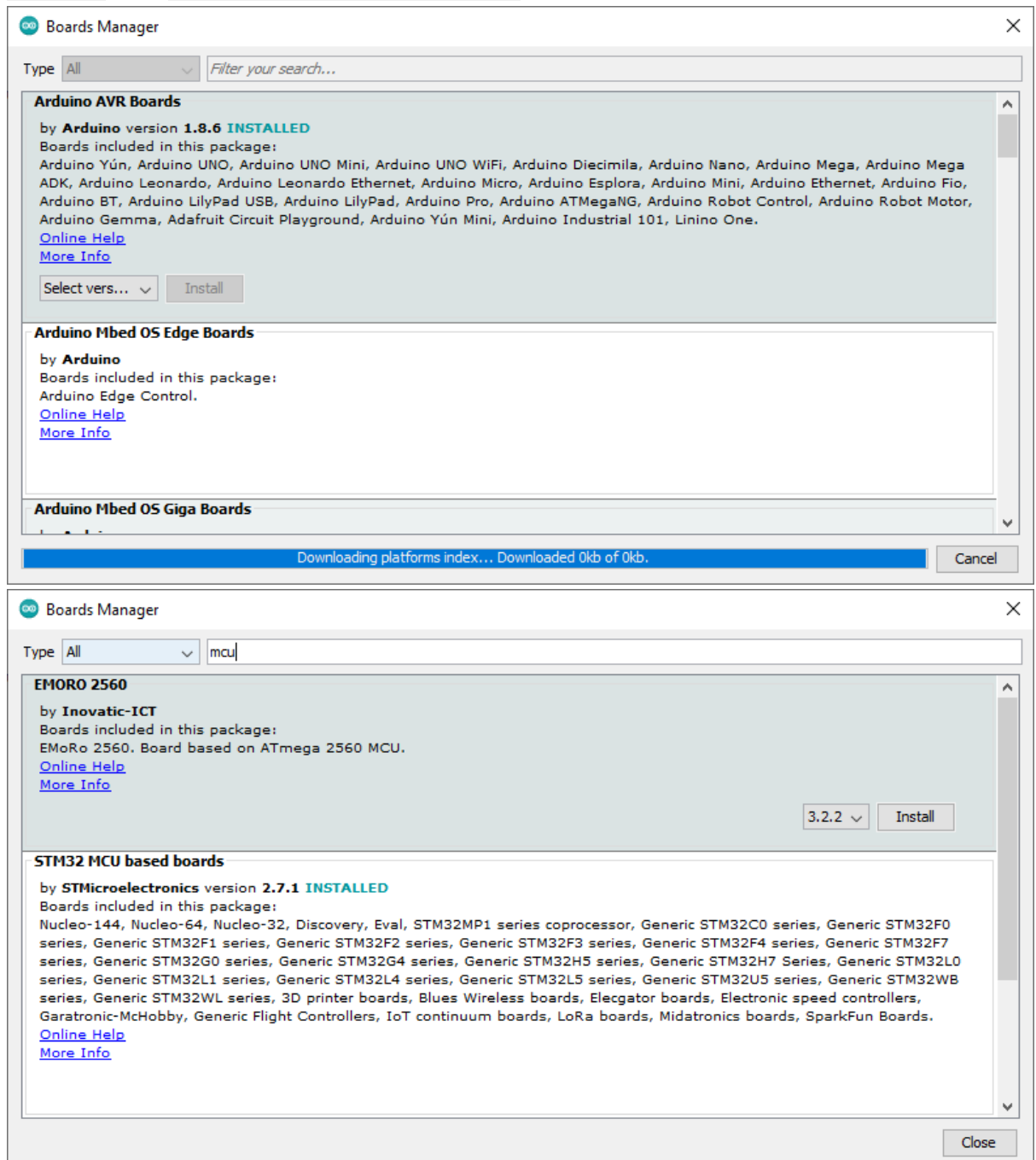
Setup Environment

Select checkboxes and allow all the permissions while installing.

1. Make sure Arduino IDE is installed.
2. Make sure the ST-Link driver is installed.
3. Make sure the STM32 Cube Programmer is installed.
4. Run Arduino IDE.
5. Open File > Preferences. Add https://raw.githubusercontent.com/stm32duino/BoardManagerFiles/main/package_stmicroelectronics_index.json to Additional Board Manager URLs. Select Ok.



6. Open Tools > Board > Boards Manager. Install Arduino AVR Boards and STM32 MCU Based Boards.



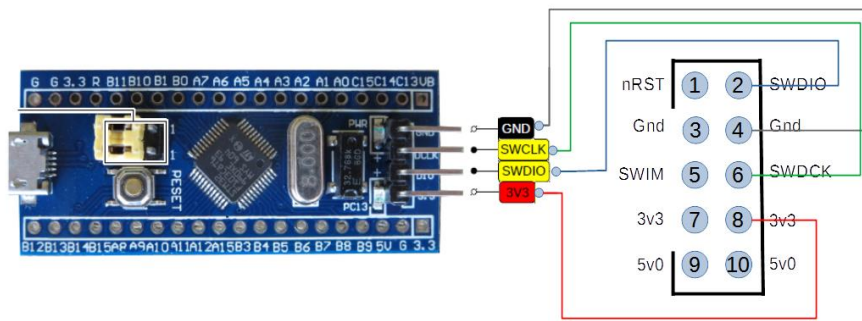
7. Connect ST-Link V2 and Blue Pill using the connector and insert ST-Link to USB. **ST-LINK V2 pinout positions may vary from tutorials. Make sure to check pinouts of your device before connecting.**

GND -> GND

3V3 -> 3V3

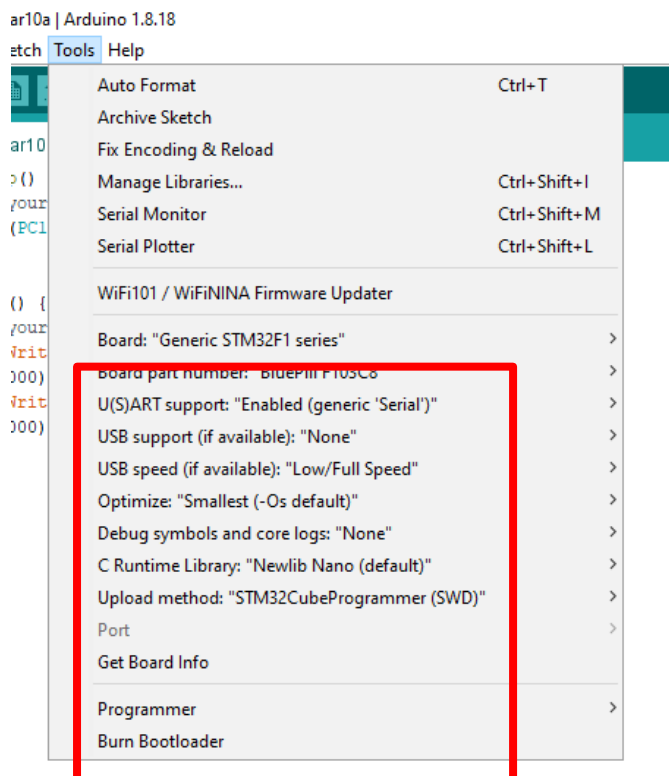
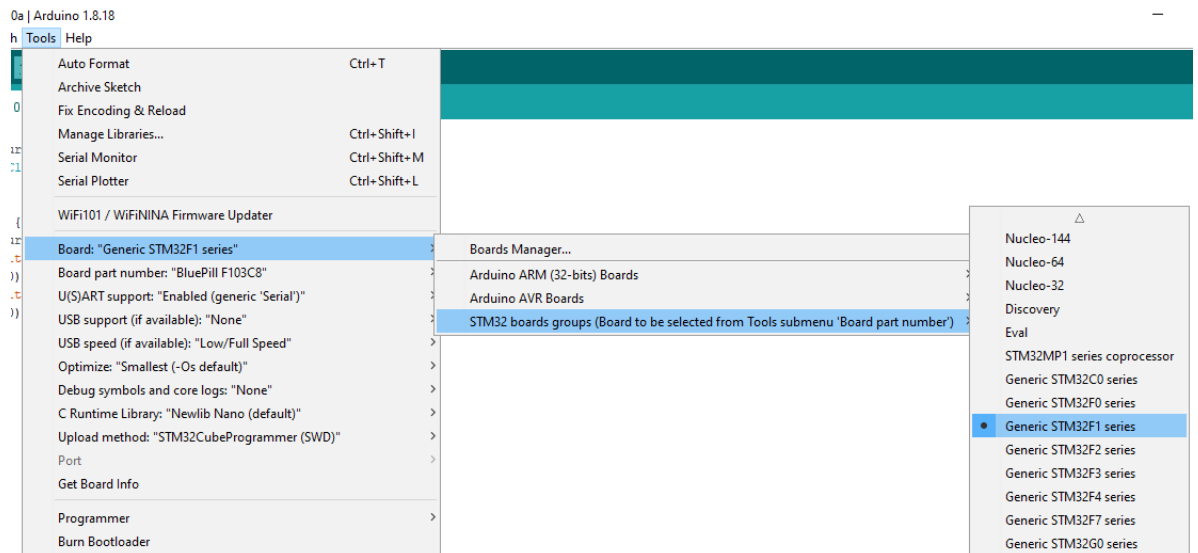
SWCLK -> SWCLK

SWIO -> SWDIO



[[Link](#)]

8. Select board and configure according to given screenshots.



9. Write your first program



```
sketch_mar10a

void setup() {
  // put your setup code here, to run once:
  pinMode(PC13, OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(PC13, HIGH);
  delay(1000);
  digitalWrite(PC13, LOW);
  delay(1000);
}
```

10. Build and Upload.



11. You should be able to see in board LED blinking.

Course Plan & Teachers

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